

Parametric Hypothesis Tests — Part 1

Basic Concepts, Test Procedure, Errors, and *p-value*

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Introduction to Hypothesis Testing

Recall: Statistical Inference

So far in this course, we have covered:

- **Sampling distributions** (Topic 1): how sample statistics behave
- **Point estimation** (Topic 2.1): finding a single “best guess” for a parameter
- **Interval estimation** (Topic 2.2): constructing a range of plausible values

Now we ask a different question:

Given a claim about a population parameter, does the sample data support or contradict that claim?

Why Hypothesis Testing?

In practice, decisions must be made under uncertainty:

- A pharmaceutical company claims a new drug lowers cholesterol by more than 20 mg/dL. Does the clinical trial data support this?
- An engineer suspects that a machine is overfilling packages beyond the nominal 100 g. Is there statistical evidence?
- A bank manager believes the average processing time for loan applications has decreased. Can we confirm this?

Hypothesis testing provides a **formal, rigorous framework** for answering these questions using sample data.

From Estimation to Testing

The Key Shift

In **estimation**, we ask: *“What is the value of the parameter?”*

In **hypothesis testing**, we ask: *“Is the parameter equal to (or greater than, or less than) a specific value?”*

We already have the building blocks:

- Sampling distributions
- Pivot (fulcral) variables: Z , T , Q (χ^2)
- Critical values from statistical tables

3.1 — Basic Concepts

The Structure of a Hypothesis Test

Every hypothesis test involves the same fundamental elements:

1. **Null hypothesis** (H_0): the claim we assume to be true until evidence says otherwise
2. **Alternative hypothesis** (H_1): the claim we are trying to find evidence for
3. **Test statistic**: a function of the sample data (computed under H_0)
4. **Significance level** (α): the probability of rejecting H_0 when it is actually true
5. **Critical (rejection) region**: the set of values of the test statistic

The Null Hypothesis (H_0)

Null Hypothesis

The **null hypothesis**, denoted H_0 , represents the *status quo* — the current belief, the manufacturer's specification, or the default assumption.

It always contains an **equality** sign ($=$, $\leq$, or $\geq$).

Examples:

- $H_0: \mu = 100$ (the mean weight is 100 g)
- $H_0: \mu \leq 500$ (the mean daily revenue does not exceed 500 €)
- $H_0: \sigma^2 \geq 10$ (the variance is at least 10)

Important: We never “accept” H_0 . We either **reject** it or **fail to reject** it.

The Alternative Hypothesis (H_1)

Alternative Hypothesis

The **alternative hypothesis**, denoted H_1 (or H_a), represents what we are trying to find evidence **for**. It is the “research hypothesis.”

It contains a strict inequality ($\neq$, $>$, or $<$).

The form of H_1 determines the **type of test**:

H_0	H_1	Type of test
$\mu = \mu_0$	$\mu \neq \mu_0$	Two-tailed (bilateral)
$\mu \leq \mu_0$	$\mu > \mu_0$	Right-tailed (unilateral right)
$\mu \geq \mu_0$	$\mu < \mu_0$	Left-tailed (unilateral left)

How to Formulate H_0 and H_1

A common source of confusion: *which claim goes in H_0 and which in H_1 ?*

Rule of thumb:

- H_0 is the claim that we assume true unless the data convinces us otherwise
- H_1 is the claim we want to **prove** (what the researcher suspects)

Think of it as a **trial**: H_0 is “innocent” (the default), and H_1 is “guilty.” The burden of proof lies on H_1 .

Example: Formulating Hypotheses

Scenario: A food company states that each package contains, on average, 500 g of product. A consumer protection agency suspects the company is underfilling.

What are H_0 and H_1 ?

- The *status quo* (company's claim): $\mu = 500$ (or $\mu \geq 500$)
- The *suspicion* (what we want evidence for): $\mu < 500$

$$H_0 : \mu \geq 500 \quad \text{vs.} \quad H_1 : \mu < 500$$

This is a **left-tailed** test.

Example: Formulating Hypotheses (cont.)

Scenario: An engineer suspects that a machine is overfilling packages beyond the nominal 100 g.

$$H_0 : \mu \leq 100 \quad \text{vs.} \quad H_1 : \mu > 100$$

This is a **right-tailed** test.

Scenario: A quality control inspector wants to check whether the mean weight differs from the specification of 100 g (could be above or below).

$$H_0 : \mu = 100 \quad \text{vs.} \quad H_1 : \mu \neq 100$$

Statistics II — Parametric Hypothesis Tests

The Test Statistic

Test Statistic

The **test statistic** is a random variable, computed from the sample, whose distribution is known **under** H_0 .

It measures *how far* the sample result is from what H_0 predicts.

You already know these from interval estimation:

Parameter	Conditions	Test Statistic	Distribution under H_0
μ	σ known	$Z_0 = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$	$N(0, 1)$
μ	σ unknown	$T_0 = \frac{\bar{X} - \mu_0}{S' / \sqrt{n}}$	$t_{(n-1)}$
σ^2	Normal pop.	$Q_0 = \frac{(n-1)S'^2}{\sigma_0^2}$	$\chi^2_{(n-1)}$

The Significance Level (α)

Significance Level

The **significance level** α is the maximum probability of rejecting H_0 when H_0 is actually true.

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

Common values: $\alpha = 0.01$, $\alpha = 0.05$, $\alpha = 0.10$.

- Smaller $\alpha \Rightarrow$ harder to reject $H_0 \Rightarrow$ stronger evidence required
- Larger $\alpha \Rightarrow$ easier to reject $H_0 \Rightarrow$ weaker evidence suffices

α is chosen **before** collecting data and performing the test.

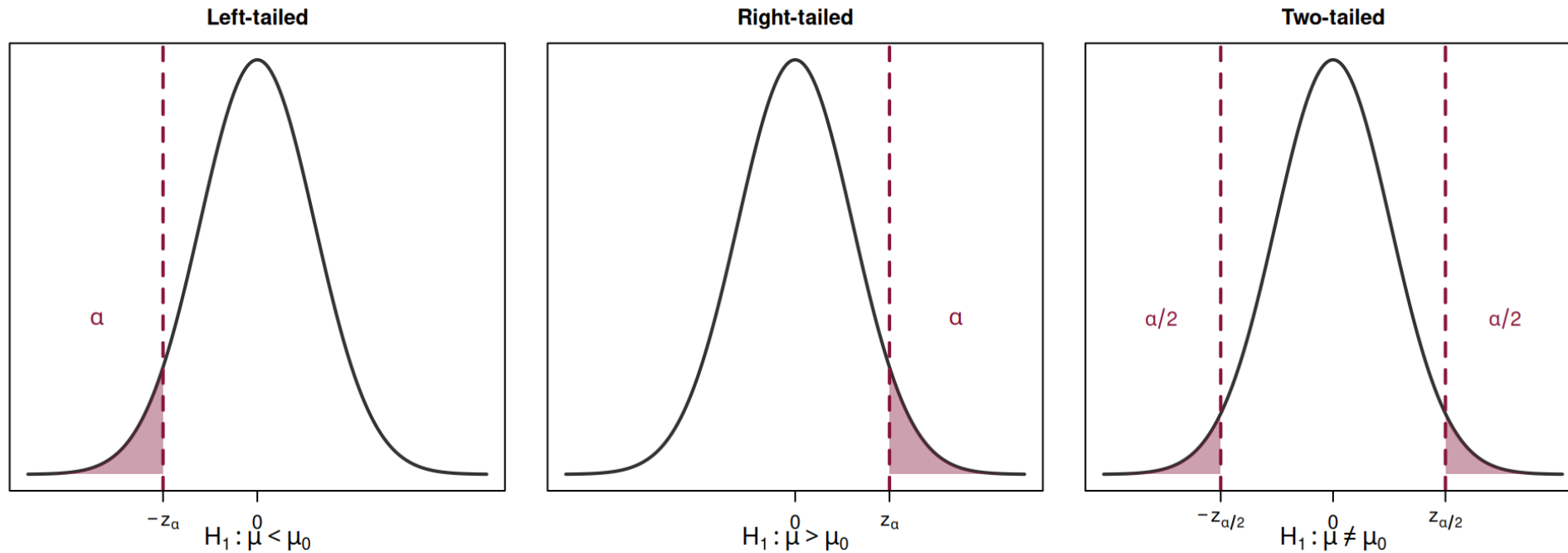
The Critical (Rejection) Region

Critical Region

The **critical region** (or rejection region) is the set of values of the test statistic for which we reject H_0 .

Its boundary is determined by the significance level α and the type of test.

Critical Regions: Visual Summary



The shaded areas represent the rejection region. If the observed test statistic falls in the shaded area, we reject H_0 .

The General Test Procedure

Step-by-step procedure for a hypothesis test

1. **State the hypotheses:** write H_0 and H_1
2. **Choose the significance level** α
3. **Identify the test statistic** and its distribution under H_0
4. **Determine the critical region** (using α and statistical tables)
5. **Compute the observed value** of the test statistic from the sample
6. **Make the decision:** reject H_0 if the observed value falls in the critical region; otherwise, do not reject H_0
7. **State the conclusion** in the context of the problem

Type I and Type II Errors

Two Types of Mistakes

When we make a decision based on sample data, we can make two kinds of errors:

	H_0 is true	H_0 is false
Do not reject H_0	Correct decision	Type II Error (β)
Reject H_0	Type I Error (α)	Correct decision

Type I Error

Type I Error

A **Type I error** occurs when we **reject** H_0 even though H_0 is **true**.

$$\alpha = P(\text{Type I Error}) = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

This is exactly the **significance level** α .

Analogy: Convicting an innocent person in a trial.

Type II Error

Type II Error

A **Type II error** occurs when we **fail to reject** H_0 even though H_0 is **false** (i.e., H_1 is true).

$$\beta = P(\text{Type II Error}) = P(\text{do not reject } H_0 \mid H_1 \text{ is true})$$

Analogy: Acquitting a guilty person in a trial.

β depends on:

- The true value of the parameter (how far from H_0)
- The sample size n
- The significance level α

The Power of a Test

Power

The **power** of a test is the probability of correctly rejecting H_0 when H_1 is true:

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ is true})$$

A good test has **high power** (close to 1).

In practice:

- Increasing n increases power (and reduces β)
- Increasing α increases power but also increases the risk of Type I error
- There is always a **trade-off** between α and β

The Trade-off Between α and β

For a **fixed sample size** n :

- If we decrease α (more conservative) $\Rightarrow \beta$ increases (harder to detect a real effect)
- If we increase α (more liberal) $\Rightarrow \beta$ decreases (easier to detect, but more false alarms)

The only way to reduce **both** α and β simultaneously is to **increase the sample size** n .

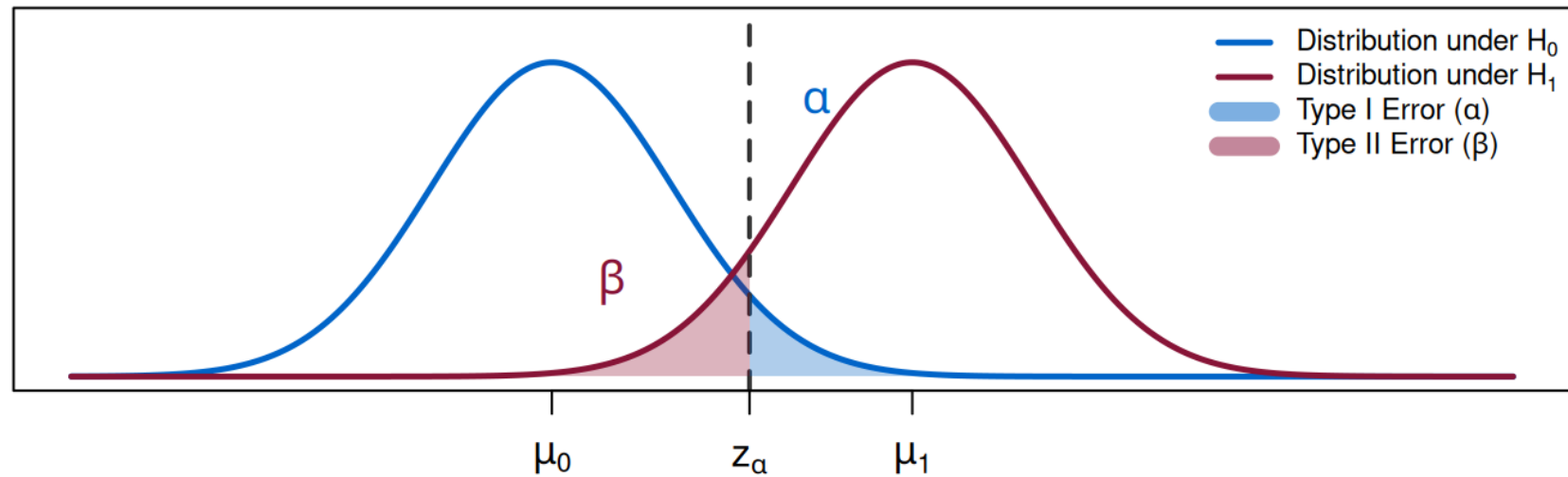
Analogy: Fire Alarm

Think of a hypothesis test as a fire alarm:

- **Type I Error (α):** The alarm goes off, but there is no fire (false alarm)
- **Type II Error (β):** There is a fire, but the alarm does not go off (missed detection)
- If we make the alarm **very sensitive**: fewer missed fires (β small), but more false alarms (α large)
- If we make the alarm **less sensitive**: fewer false alarms (α small), but more missed fires (β large)

Summary: Errors at a Glance

Type I and Type II Errors



Solved Example — Putting It All Together

Example 1: Test for the Mean (σ known)

A bottling company fills bottles with a nominal volume of $\mu_0 = 500$ mL. The filling process is known to have a standard deviation of $\sigma = 8$ mL. A quality inspector takes a random sample of $n = 36$ bottles and obtains a sample mean of $\bar{x} = 497$ mL.

At the 5% significance level, is there evidence that the machine is underfilling?

Example 1: Solution — Step 1

Step 1: State the hypotheses.

The inspector suspects underfilling, i.e., the mean is *less than* 500.

$$H_0 : \mu \geq 500 \quad \text{vs.} \quad H_1 : \mu < 500$$

This is a **left-tailed** test.

Example 1: Solution — Step 2

Step 2: Significance level.

$$\alpha = 0.05$$

Example 1: Solution — Step 3

Step 3: Test statistic.

The population standard deviation $\sigma = 8$ is **known**, so:

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \underset{\text{under } H_0}{\sim} N(0, 1)$$

Example 1: Solution — Step 4

Step 4: Critical region.

For a left-tailed test at $\alpha = 0.05$:

$$\text{Rejection region: }]-\infty ; -z_{\alpha}] =]-\infty ; -1.645]$$

We reject H_0 if $z_{obs} \leq -1.645$.

Example 1: Solution — Step 5

Step 5: Compute the observed value.

$$z_{obs} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{497 - 500}{8 / \sqrt{36}} = \frac{-3}{8/6} = \frac{-3}{1.333} \approx -2.25$$

Example 1: Solution — Step 6

Step 6: Decision.

$$z_{obs} = -2.25 \leq -1.645$$

The observed value **falls in the rejection region**.

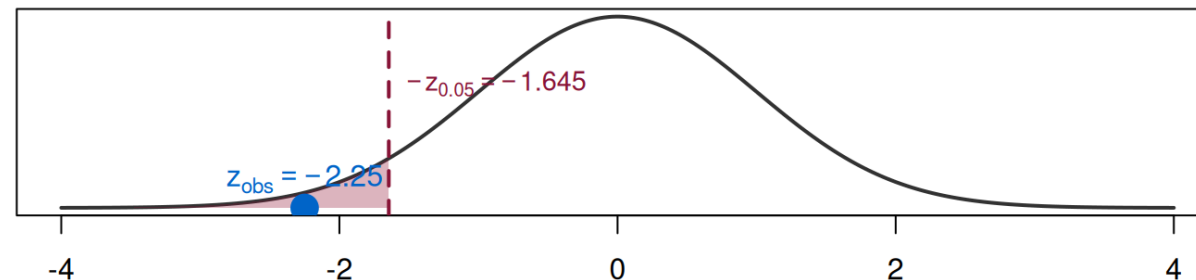
⇒ We **reject** H_0 at the 5% significance level.

Example 1: Solution — Step 7

Step 7: Conclusion.

At the 5% significance level, there is **sufficient statistical evidence** to conclude that the machine is underfilling the bottles (i.e., the mean volume is less than 500 mL).

Left-tailed test: rejection region and observed value



Example 2: Formulating Hypotheses — Practice

For each scenario, write H_0 and H_1 , and state the type of test:

- a. A bank claims the average time to process a loan is 5 days. A consultant believes it takes longer.
- b. A manufacturer states the defect rate is at most 3%. An auditor suspects it is higher.
- c. A researcher wants to test if a training program changes employee productivity from the current average of 40 units/day.

Example 2: Solutions

a. $H_0 : \mu \leq 5$ vs. $H_1 : \mu > 5 \rightarrow$ Right-tailed

b. $H_0 : p \leq 0.03$ vs. $H_1 : p > 0.03 \rightarrow$ Right-tailed

c. $H_0 : \mu = 40$ vs. $H_1 : \mu \neq 40 \rightarrow$ Two-tailed

Note: In (c), the researcher does not have a directional suspicion — it could go either way — hence the two-tailed test.

Quick Recap Before the Break

What we covered so far

- **Null hypothesis** (H_0): status quo, contains $=$, $\leq$, or $\geq$
- **Alternative hypothesis** (H_1): research claim, contains $\neq$, $>$, or $<$
- **Test statistic**: pivot variable computed under H_0
- **Significance level** (α): $P(\text{reject } H_0 \mid H_0 \text{ true})$
- **Critical region**: values that lead to rejection of H_0
- **Type I Error** (α): rejecting a true H_0
- **Type II Error** (β): not rejecting a false H_0
- **Power** $= 1 - \beta$

3.2 — The p -value

Motivation

Before the break, we used the **critical region** approach:

1. Fix α
2. Find the critical value(s)
3. Compare z_{obs} (or t_{obs} , q_{obs}) with the critical value

But this only tells us “reject” or “do not reject” for a **specific** α .

What if someone asks: “*Would you still reject H_0 at $\alpha = 0.01$?*”

The **p -value** answers this question for **all possible** significance levels at once.

Definition of the p -value

p -value

The p -value is the probability, computed under H_0 , of observing a test statistic value **as extreme as, or more extreme than**, the value actually observed.

It is the **smallest significance level** at which we would reject H_0 .

Informally: the p -value measures *how surprised we should be* by the sample data, if H_0 were true.

- Small p -value $\Rightarrow$ data is very unlikely under $H_0 \Rightarrow$ strong evidence against H_0
- Large p -value $\Rightarrow$ data is consistent with $H_0 \Rightarrow$ no evidence against H_0

p-value Formulas

The formula depends on the type of test:

Type of test	<i>p</i> -value
Left-tailed ($H_1 : \theta < \theta_0$)	$p = P(T \leq t_{obs} \mid H_0)$
Right-tailed ($H_1 : \theta > \theta_0$)	$p = P(T \geq t_{obs} \mid H_0)$
Two-tailed ($H_1 : \theta \neq \theta_0$)	$p = 2 \times P(T \geq t_{obs} \mid H_0)$

Here T denotes the test statistic (could be Z , T , Q , etc.) and t_{obs} is its observed value.

Decision Rule Using the p -value

p -value decision rule

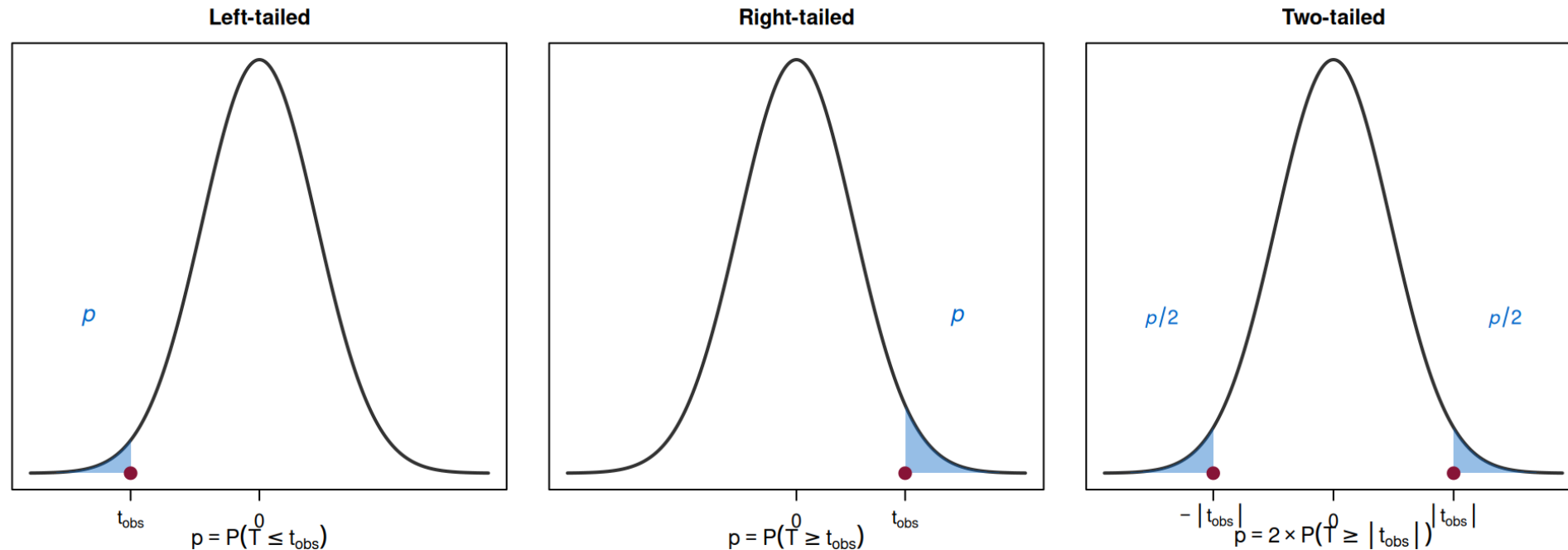
Given a significance level α :

- If $p\text{-value} \leq \alpha \Rightarrow$ **reject** H_0
- If $p\text{-value} > \alpha \Rightarrow$ **do not reject** H_0

This is equivalent to the critical region approach, but more informative:

- The critical region approach gives a binary answer for one specific α
- The p -value tells you **exactly how much evidence** the data provides against H_0

p -value: Graphical Interpretation



The shaded blue areas represent the p -value: the probability of obtaining a result as extreme as t_{obs} , under H_0 .

Example 3: p -value Calculation (cont. from Example 1)

Recall: $H_0 : \mu \geq 500$ vs. $H_1 : \mu < 500$, $z_{obs} = -2.25$, left-tailed test.

$$\begin{aligned} p\text{-value} &= P(Z \leq z_{obs} \mid H_0) = P(Z \leq -2.25) \\ &= \Phi(-2.25) = 1 - \Phi(2.25) = 1 - 0.9878 = 0.0122 \end{aligned}$$

Interpretation: If H_0 were true ($\mu = 500$), the probability of observing a sample mean as low as (or lower than) 497 is only 1.22%.

Decision (at $\alpha = 0.05$): Since $p = 0.0122 < 0.05 = \alpha$, we reject H_0 .

Decision (at $\alpha = 0.01$): Since $p = 0.0122 > 0.01 = \alpha$, we do **not** reject H_0 .

Example 4: p -value with the t -distribution

A random sample of $n = 20$ observations is drawn from a Normal population. The sample mean is $\bar{x} = 101$ and the corrected sample standard deviation is $s' = 3$.

Test $H_0 : \mu \leq 100$ vs. $H_1 : \mu > 100$.

Example 4: Solution

Test statistic (σ unknown, Normal population):

$$T_0 = \frac{\bar{X} - \mu_0}{S' / \sqrt{n}} \underset{\text{under } H_0}{\sim} t_{(n-1)} = t_{(19)}$$

Observed value:

$$t_{obs} = \frac{101 - 100}{3 / \sqrt{20}} = \frac{1}{3 / 4.472} = \frac{1}{0.6708} \approx 1.49$$

Example 4: Solution (cont.)

p -value (right-tailed test):

$$p = P(T > 1.49 \mid H_0)$$

Using the t -Student table (Table 7) with 19 degrees of freedom:

We look for the row $\nu = 19$. We find that $t_{0.10,(19)} = 1.328$ and $t_{0.05,(19)} = 1.729$.

Since $1.328 < 1.49 < 1.729$, we have:

$$0.05 < p < 0.10$$

Decision (at $\alpha = 0.05$): Since $p > 0.05$, we **do not reject** H_0 .

Statistics II — Parametric Hypothesis Tests

Conclusion: At the 50% level, there is **not sufficient evidence** to

Reading the p -value from Tables

When using the t -Student or χ^2 tables, we typically cannot compute the exact p -value. Instead, we **bracket** it:

Strategy: Find the two table entries that “sandwich” t_{obs} (or q_{obs}).

Example: $t_{obs} = 2.861$, $\nu = 19$ (right-tailed).

From Table 7: $t_{0.005, (19)} = 2.861$.

So $p = 0.005$ (exactly, in this case).

Tip: If t_{obs} matches a table entry exactly, the p -value is the corresponding tail probability. Otherwise, state the interval.

Example 5: Two-tailed p -value

A manufacturer specifies that the mean diameter of a component is $\mu_0 = 25$ mm. The population variance is known: $\sigma^2 = 4$. A random sample of $n = 49$ components has $\bar{x} = 25.6$ mm.

Test $H_0 : \mu = 25$ vs. $H_1 : \mu \neq 25$. Compute the p -value.

Example 5: Solution

Test statistic (σ known):

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Observed value:

$$z_{obs} = \frac{25.6 - 25}{2/\sqrt{49}} = \frac{0.6}{2/7} = \frac{0.6}{0.2857} \approx 2.10$$

p -value (two-tailed):

$$\begin{aligned} p &= 2 \times P(Z \geq |z_{obs}|) = 2 \times P(Z \geq 2.10) \\ &= 2 \times [1 - \Phi(2.10)] = 2 \times (1 - 0.9821) = 2 \times 0.0179 = 0.0358 \end{aligned}$$

Decision: $p = 0.0358 < 0.05$, so we reject H_0 at $\alpha = 0.05$. There is evidence that $\mu \neq 25$ mm.

Equivalence: Critical Region vs. p -value

Both approaches always give the **same conclusion** for a given α :

Approach	Reject H_0 if...
Critical region	t_{obs} falls in the rejection region
p -value	$p\text{-value} \leq \alpha$

In practice, many practitioners prefer the p -value because:

- It provides **more information** (the exact strength of evidence)
- It allows the reader to draw their own conclusion for their preferred α

- It is the standard output of statistical software

Common Misinterpretations of the p -value

The p -value is NOT:

- The probability that H_0 is true
- The probability that H_1 is false
- The probability of making an error

The p -value IS:

The probability of observing data as extreme as (or more extreme than) what was observed, **assuming H_0 is true**.

Practical Guidelines for Interpreting p -values

While the decision rule ($p \leq \alpha \Rightarrow \text{reject}$) is clear-cut, some authors provide informal guidelines:

p -value	Evidence against H_0
$p > 0.10$	No evidence
$0.05 < p \leq 0.10$	Weak (marginal) evidence
$0.01 < p \leq 0.05$	Moderate evidence
$0.001 < p \leq 0.01$	Strong evidence
$p \leq 0.001$	Very strong evidence

Source: adapted from Newbold, Carlson & Thorne.

These are informal guidelines, not strict rules. The choice of α remains a decision of the researcher.

Relationship Between Confidence Intervals and Hypothesis Tests

The Connection

There is a direct relationship between a **two-tailed** hypothesis test and a **confidence interval**:

CI-Test Equivalence

At significance level α , we **reject** $H_0 : \mu = \mu_0$ (two-tailed) if and only if μ_0 **does not belong** to the $(1 - \alpha) \times 100\%$ confidence interval for μ .

This makes intuitive sense: if the hypothesized value μ_0 is outside the plausible range given by the CI, then the data contradicts H_0 .

Example 6: CI and Test Equivalence

Recall Example 5: $\mu_0 = 25$, $\sigma = 2$, $n = 49$, $\bar{x} = 25.6$.

The 95% confidence interval for μ is:

$$\bar{x} \pm z_{0.025} \cdot \frac{\sigma}{\sqrt{n}} = 25.6 \pm 1.96 \times \frac{2}{7} = 25.6 \pm 0.56$$

$$\text{CI} = [25.04 ; 26.16]$$

Since $\mu_0 = 25 \notin [25.04 ; 26.16]$, we reject $H_0 : \mu = 25$ at $\alpha = 0.05$.

This is consistent with our earlier finding ($p = 0.0358 < 0.05$).

Solved Exercises

Exercise 1

A manufacturer claims that light bulbs last, on average, at least 1000 hours. A random sample of 25 bulbs from a Normal population gives $\bar{x} = 980$ hours and $s' = 40$ hours.

At $\alpha = 0.05$, is there evidence against the manufacturer's claim?

Exercise 1: Solution

Step 1: $H_0 : \mu \geq 1000$ vs. $H_1 : \mu < 1000$ (left-tailed)

Step 2: $\alpha = 0.05$

Step 3: σ unknown, Normal population, so:

$$T_0 = \frac{\bar{X} - \mu_0}{S' / \sqrt{n}} \underset{\text{under } H_0}{\sim} t_{(24)}$$

Exercise 1: Solution (cont.)

Step 4: Left-tailed, $\alpha = 0.05$, $\nu = 24$

$$\text{Rejection region: }]-\infty ; -t_{0.05,(24)}] =]-\infty ; -1.711]$$

(From Table 7: $t_{0.05,(24)} = 1.711$)

Step 5:

$$t_{obs} = \frac{980 - 1000}{40/\sqrt{25}} = \frac{-20}{8} = -2.50$$

Step 6: $t_{obs} = -2.50 < -1.711 \rightarrow$ falls in the rejection region.

Decision: Reject H_0 .

Exercise 1: p -value

$$p = P(T \leq -2.50) = P(T \geq 2.50)$$

From Table 7, $\nu = 24$: $t_{0.01,(24)} = 2.492$ and $t_{0.005,(24)} = 2.797$.

Since $2.492 < 2.50 < 2.797$, we have $0.005 < p < 0.01$.

Conclusion: At 5%, there is strong evidence that the mean lifetime is less than 1000 hours ($p < 0.01$).

Exercise 2

A food processing company fills cereal boxes labeled as 500 g. The variance of the filling process is known to be $\sigma^2 = 25 \text{ g}^2$. A random sample of $n = 64$ boxes yields $\bar{x} = 498 \text{ g}$.

- a. Test whether the mean weight differs from 500 g at $\alpha = 0.05$.
- b. Compute the p -value.

Exercise 2: Solution

a) $H_0 : \mu = 500$ vs. $H_1 : \mu \neq 500$ (two-tailed)

$\sigma = 5$ is known, $n = 64$:

$$Z_0 = \frac{\bar{X} - 500}{\sigma/\sqrt{n}} \sim N(0, 1) \quad z_{obs} = \frac{498 - 500}{5/\sqrt{64}} = \frac{-2}{5/8} = \frac{-2}{0.625} = -3.20$$

Rejection region (two-tailed, $\alpha = 0.05$):

$$]-\infty; -z_{0.025}] \cup [z_{0.025}; +\infty[=]-\infty; -1.96] \cup [1.96; +\infty[$$

Since $z_{obs} = -3.20 < -1.96$, we **reject** H_0 .

Exercise 2: Solution (cont.)

b) p -value (two-tailed):

$$p = 2 \times P(Z \geq |-3.20|) = 2 \times P(Z \geq 3.20)$$

$$= 2 \times [1 - \Phi(3.20)] = 2 \times (1 - 0.9993) = 2 \times 0.0007 = 0.0014$$

Very strong evidence against H_0 : the p -value is much smaller than any conventional α .

Exercise 3

From a random sample of $n = 20$ observations drawn from a Normal population, one obtains $\bar{x} = 101$ and $s' = 3$.

Consider the test: $H_0 : \mu \leq 100$ vs. $H_1 : \mu > 100$.

- a. For $\alpha = 0.05$, determine the rejection region.
- b. If, for another sample of the same size, $t_{obs} = 2.861$, what is the p -value?

Exercise 3: Solution

a) Right-tailed test, $T_0 \sim t_{(19)}$:

$$\text{Rejection region: } [t_{0.05,(19)} ; +\infty[= [1.729 ; +\infty[$$

(Table 7: $t_{0.05,(19)} = 1.729$)

b) For $t_{obs} = 2.861$, right-tailed:

$$p = P(T > 2.861)$$

From Table 7, $\nu = 19$: $t_{0.005,(19)} = 2.861$.

So $p = 0.005$.

There is evidence to reject H_0 for all usual significance levels ($0.005 < \alpha$ for any conventional α)

Summary of Today's Lecture

Key takeaways — Lecture 6

1. A hypothesis test is a formal procedure to evaluate a claim about a population parameter
2. H_0 (status quo) vs. H_1 (research hypothesis); H_0 always contains equality
3. The **test statistic** is a pivot variable computed under H_0
4. **Type I Error** (α) = rejecting true H_0 ; **Type II Error** (β) = not rejecting false H_0
5. The **p-value** is the smallest α at which we would reject H_0
6. Decision: reject H_0 if $p \leq \alpha$, or equivalently, if t_{obs} falls in the rejection region
7. A two-tailed test at level α and a $(1 - \alpha)$ CI give equivalent conclusions

Next Lecture (April 21)

We will apply the testing framework to specific cases:

- **Section 3.3:** Tests for means and variances of Normal populations
- **Section 3.4:** Tests for the difference of means of two Normal populations
- **Section 3.5:** Large-sample tests (asymptotic normality)

Make sure to review the relevant sections in Newbold (Chapter 9).

Disclaimer

These slides are a free adaptation of the course material for Estatística II by Prof. Teresa Ferreira and Prof. Sandra Custódio from the Lisbon Accounting and Business School — Polytechnic University of Lisbon.

Primary reference: Newbold, P., Carlson, W. & Thorne, B. — *Statistics for Business and Economics*, Global Edition.